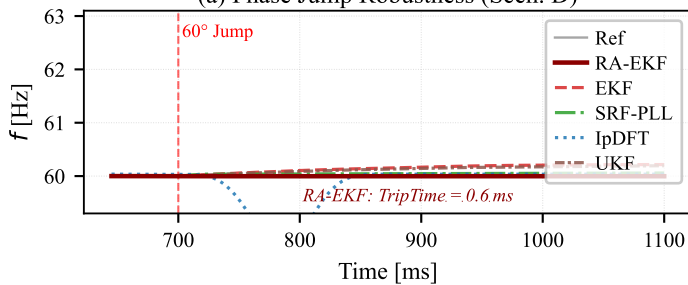
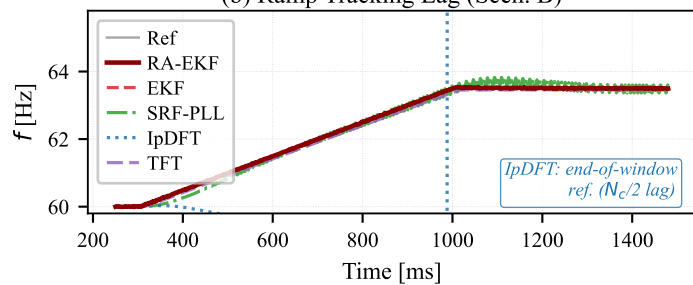


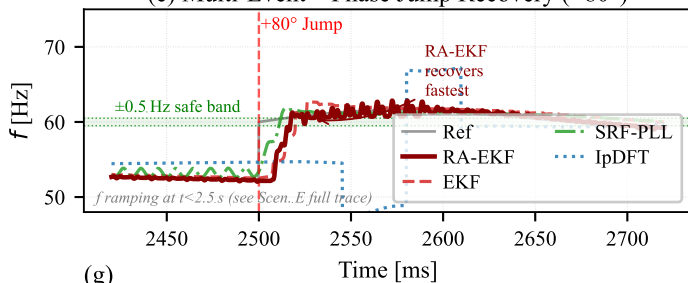
(a) Phase Jump Robustness (Scen. D)



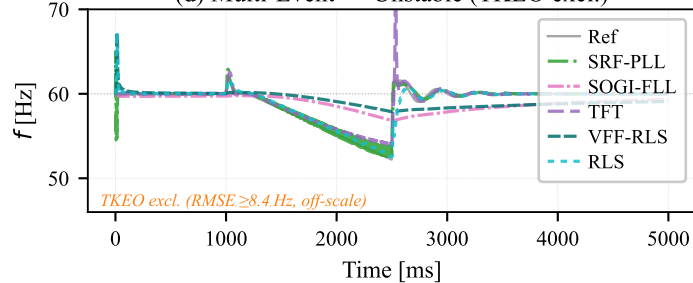
(b) Ramp Tracking Lag (Scen. B)



(c) Multi-Event—Phase Jump Recovery (+80°)



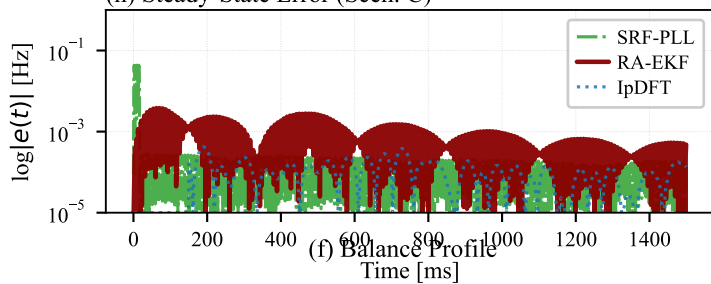
(d) Multi-Event — Unstable (TKEO excl.)



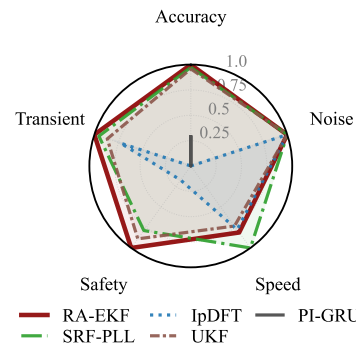
(g)

	Step	Ramp	Mod	Isl	Multi
PI-GRU	F	F	F	F	F
Koopman	P	F	P	F	F
TFT	P	F	P	F	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	F	P	P	F
SRF-PLL	P	F	P	P	F
EKF	P	P	P	F	F
UKF	P	P	P	F	F
RA-EKF	P	P	P	P	F

(h) Steady-State Error (Scen. C)



(f) Balance Profile



(e) Cost vs. Risk [Multi-Event]

